

Stock Price Prediction Using Machine Learning Models

1. Introduction

Large amounts of time-series data are produced by financial markets, which can be used to create trading and risk management prediction models. Designing an end-to-end data science pipeline to evaluate past stock price data and create predictive models for stock price forecasting is the project's goal. This project uses daily OHLC and volume data from publicly accessible market feeds to compare a machine-learning technique (Gradient Boosting Regressor) with a conventional time-series model (ARIMA).

2. Dataset

The yfinance Python library was used to retrieve daily historical stock price data from Yahoo Finance. The following stocks are included in the dataset:

- Apple (AAPL)
- Microsoft (MSFT)
- Google (GOOG)
- Amazon (AMZN)
- Tesla (TSLA)

Every dataset includes:

- Open
- High
- Low
- Close
- Adjusted Close
- Volume

While other stocks are used for exploratory analysis, Apple (AAPL) is the main focus of the modeling and analysis for demonstration.

3. Data Preparation and Cleaning

We needed to clean up the raw data a bit before we could use it for our time-series modeling. Here's what I did:

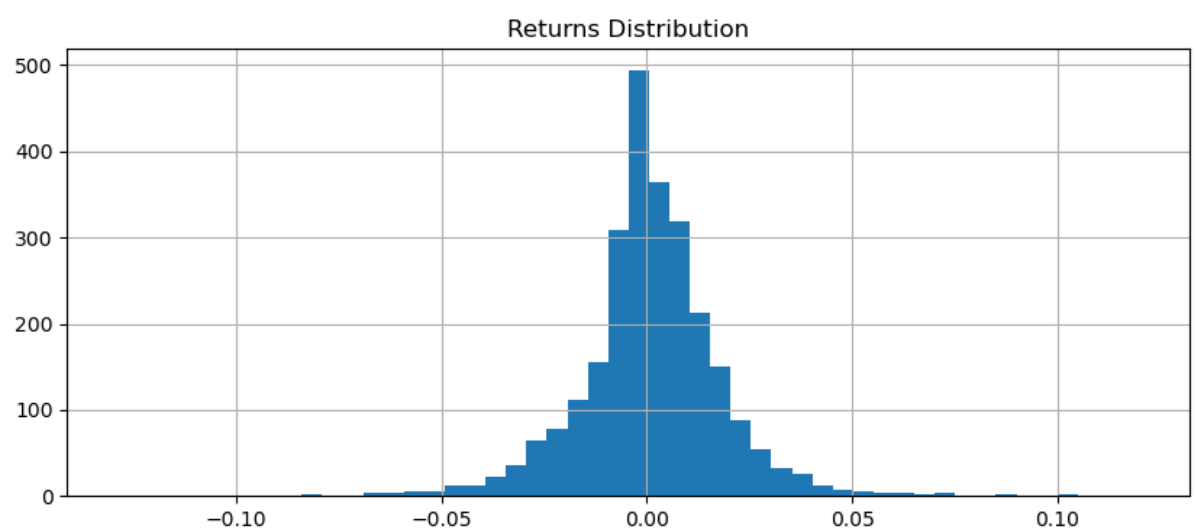
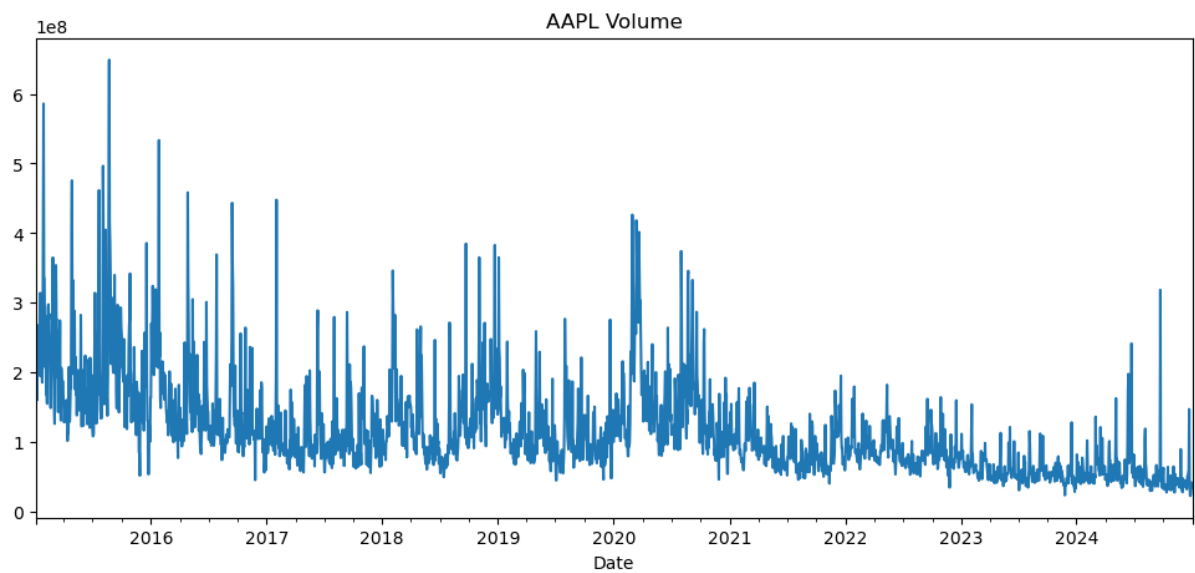
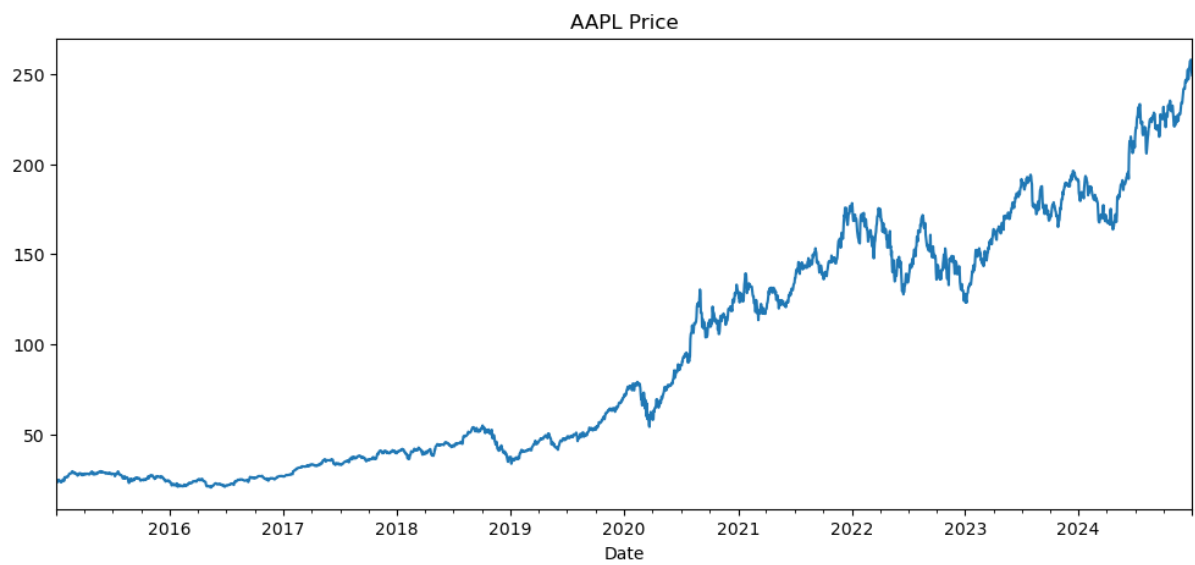
- put the observations in order from oldest to newest.
- got rid of the extra timestamps.
- filled in the blank spots with the last known good value.
- made sure that none of the prices or amounts were negative.
- put the data back in order, but this time I made sure it only included business days. This way, it plays nice with all our time-series models.
- filled the gaps for all the business days missed because of weekends and holidays.

4. Exploratory Data Analysis (EDA)

I took a look at past prices and how the market acted to get a better sense of things.

Important observations:

- Over time, stock prices generally go up, but they also have their ups and downs.
- A lot more trading is seen whenever something big happens in the market.
- Every day, stock returns pretty much hover around zero, but every now and then, you see some really wild swings.
- Rolling averages sure do help smooth out those up-and-down changes, making trends a lot easier to spot.



5. Feature Engineering

To enhance the model's predictive power, additional variables were formulated using the original price data. Additionally, lagged price variables are used in this model by shifting the data two time units into the past. These variables include the 'price last day' and 'price last 5 days.' These variables attempt to extract short-term as well as long-term dependencies in the data. Returns are given as percentage changes over the past one day and five days. Moving averages defined as the mean values over the past 5 days and 20 days. Volatility defined as the standard deviation over the last 20 days.

6. Models Used

6.1 ARIMA Model

The ARIMA model, or AutoRegressive Integrated Moving Average model, was employed as a baseline time-series forecasting method. The model exploits linear relationships within the time-series price data, assuming that the previous price movements contain information regarding future price movements.

Steps:

- Enforced Stationarity using Differencing
- This ARIMA model was selected based on empirical testing.
- Trained with 80% of the historical data, tested on remaining 20%.

Although ARIMA offers interpretable results, it has limitations in processing non-linear relationships.



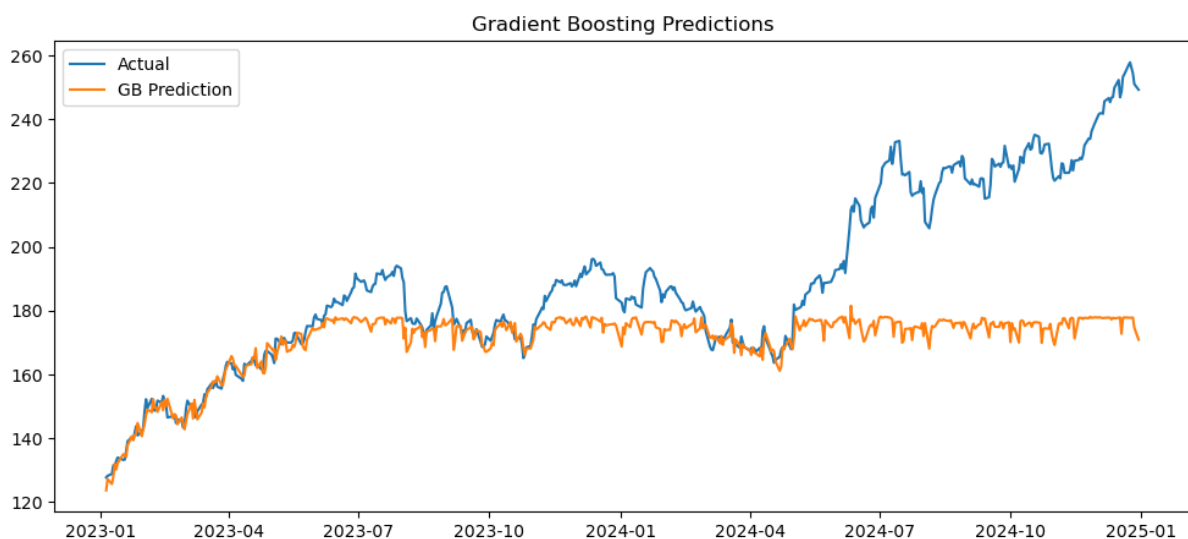
6.2 Gradient Boosting Regressor

A Gradient Boost Regressor model has been employed to detect the non-linear relationships between the features and future variables.

Key characteristics:

- Uses ensemble decision trees
- Utilizes multiple simultaneous explanatory signals
- Trained using time-based train-test split to avoid data leakage

The hyperparameters, such as the learning rate, tree depth, and number of estimators, were used.



7. Model Evaluation

Both models were evaluated using standard regression metrics:

- RMSE
- MAE
- MAPE

Model	RMSE	MAE	MAPE
ARIMA	66.778268	60.572424	30.652214
Gradient Boosting	28.505086	18.670444	8.644381

The Gradient Boosting model performed better than the ARIMA model in all metrics, signifying better predictive accuracy.

8. Trading Implications

From a trading perspective:

- ARIMA can be more appropriate for trend following in stable regimes
- Gradient Boosting better handles short-term price behavior.
- Neither model should be employed independently as a trading system
- Predictions can be integrated as signals within the risk management and execution logic

These models are most appropriate for signal confirmation, position sizing, or risk assessment rather than for actual trading.

9. Limitations and Future Work

Limitations:

- Limited number of equities
- No macroeconomic or sentiment data
- Single-step forecasting horizon

Future improvements:

- Adding more assets and their histories

- Use walk-forward validation
- Explore LSTM, XGBoost, or Transformer-based models
- Integrate Regime Detection and Volatility Filters

10. Conclusion

This project shows the application of the data science pipeline to the prediction of the stock exchange price using a combination of conventional methods and machine learning. It has been noted that the results of machine learning models using engineered features are more accurate than conventional ARIMA models. This project may be the basis of more complex quantitative models.